Rerunning every simulation affected by the parameter error: plan for four hosts

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1. Purpose and scope

This plan reruns every simulation whose reported results rest on a parameter set other than Hendrickson et al.'s published values (`docs/70`): the 'simcore2026' set of `simulation-core.R` and the 'raw fixture' copied from the unit tests. It extends `docs/73` (papers 01 and 02 only) to every affected paper and adds two hosts, athena and joe, which became simrun hosts on 2026-11-03. Where the two plans differ, this one governs.

The inventory comes from four read-only audits on 2026-11-03, which read each driver's parameter source, the files each manuscript reads, and the stored runs' provenance. Their findings are summarized in Sections 3 and 4; each statement there is marked verified (read in code, provenance or a stored file) or inferred.

Affected and in this plan: papers 01, 02, 03, 06, 08, 09 and 10; the cr2-olbdc check (read by papers 02 and 09); and the prognostic material leaving paper 02 for a companion paper (paper 15, whose folder does not exist yet).

Confirmed unaffected (verified): papers 04 and 05 use their own, stated parameter sets; papers 07, 11, 13 and 14 load the published values from the package data; paper 12 inherits published values from an archived paper 11 run; paper 01's closed form, study 2 and study 3 use the published values.

2. Hosts and measured capacity

Benchmarks on 2026-11-03 ran driver 19 at commit c866404 in the project image (`...:10504a44fb8d`) on each host.

Host	Hardware	One worker, heaviest cell (s per replicate)	8-worker block F run	Throughput relative to zeus
zeus	Linux server, 12 threads	2.05	6.5 min	1.0
athena	M1 Max, 10 cores, Docker 24 GB	5.57	8.2 min	about 0.8
joe	M2, 8 cores, Docker 20 GB	4.90	not run	about 0.5 (inferred)
julia (overnight only)	M1, 8 cores, Docker 12 GB	5.47	not run	about 0.4 (inferred)
mercury	Linux, 6 cores used	not run	not run	about 0.4 (job 40)

- The Macs run the amd64 image under Rosetta, about 2.6 times slower per worker than zeus, but athena keeps about two thirds of its single-worker speed at 8 workers, against about a third on zeus. One maths-library thread per worker did not change zeus's result (7.4 against 6.5 minutes), so the cause of zeus's poor scaling is unknown.
- An earlier inference, in `docs/73` and `docs/72`, that zeus runs at a quarter of julia's per-core speed was wrong: one zeus worker matches julia natively on light cells and is about 35 percent slower on the heaviest. The fourfold gap is a loss when 8 workers run together.
- **Pairing across hosts.** Results are not draw-for-draw identical between hosts, and the Macs' emulated runs are not identical to zeus's. Jobs that must be compared trial by trial run on the same host (Section 6).
- **Memory.** athena's Docker Desktop has 24 GB (jobs use 22) and joe's 20 GB (jobs use 18), both from 2026-11-03.

3. Inventory by paper

Hours are wall-clock on zeus at 8 cores ('zeus-hours') where a stored zeus job did the same work, and local hours on julia (native, about 7 workers) where only a local run exists; local hours are not converted, because the relation between julia's native multi-worker speed and the container hosts was not measured.

Ready now: the parameter switch reaches the driver

Paper	Job	Driver	Cost	Status
01	p01-pf	19, blocks P and F, G10 and G9	11.9 zeus-h	ready (`docs/73`)
01	p01-n80	19, block N	4.7 zeus-h	ready
02	ten jobs	01, 27, 38, 06, 32 (three), 36 (three)	33.0 zeus-h	ready (`docs/73` Section 6)
15	p15-gapC	32 `--block C`	about 13.5 h on mercury (job 40) plus G10	ready; needs a new `--out-dir` (verified)
15	p15-refT	36 `--block T`	0.13 zeus-h (job 36)	ready
15	(c_pb = 0.3 cells)	36 `--block N`	in paper 02's ref-N job	no extra job (verified)

Small code changes first (Tier B)

Paper	Job	Driver	Cost	Change needed
10	p10	`run-paper10-reduced.sh` (nine drivers)	1.8 zeus-h (job 55)	`p10_out_dir()` and the wrapper write to a folder per parameter set (`output/paper10-published/`), record `param_set()` in metadata; the Rmd reads that folder; export `PMSIM_PARAMS=published` in the wrapper. The kappa script's printed 'sigma_u = 10' and Rmd line 720 ('B_i drawn from N(0, 1)') change.
15	p15-stress	28	5.5 zeus-h (job 28)	add `guard_out_dir()` and `param_meta()`; new `--out-dir` (it resumes from `trials/` checkpoints)
15	p15-re	29	3.2 zeus-h (job 30)	same as 28; same host as 28 (shared seeds)

Paper	Job	Driver	Cost	Change needed
03	p03-null	`12-referee-rerun.R`; null block	20 h, us-h (job 38)	add `guard_out_dir()` and `param_meta()`; new out-dir; standardize `bm` before `lme_analysis()` and `hlme(classmb = ~bm)` (they use the raw biomarker)
03	p03-alt	same, alternative block	unknown: job 39 was cancelled after 152 h	as p03-null; run `--time` first and size from it
02, 09	cr2-p02	`cr2-olbdc-check.out-run-cr2-null.R`, p02 setting	4 h, local h	add `PMSIM_PARAMS=published` (its wrapper passes no `--params`); new `CHK_OUT` (the summarizer pools every file in the folder); the p02 baseline rows use the raw biomarker

Larger code changes first (Tier C)

These drivers hard-code the raw fixture (TV and BR SD 5, PB SD 2, the TV mean on the BR curve, biomarker 0/1, baseline 70/10; verified) and do not source `simulation-core.R`. Each needs a parameter switch, a true value taken from the active set, a biomarker scale decision (Section 5, D1), an output folder that cannot overwrite the stored run, and worker sizing from `SIM_CORES`.

Paper	Job	Driver	Stored cost	Notes
06	p06-s1	`run-study-a-procedure.R` (Study 1)	about 13 local h (7 workers)	true value hard-coded at -2.25 and typed in the text; the m_PB and m_TV axes weaken against published SDs (D6)
06	p06-s2	`run-study-2-feasible.R` (corrected Study 2)	43 local h (job 49; ran 11 workers on an 8-core cgroup)	contamination bias grows about 3 times relative to the signal under published SDs (inferred)
06	p06-s4	`run-study-b-balanced.R` (Study 4)	about 0.7 local h	`TRUE_BETA` hard-coded
06	p06-bounds	`05-feasibility-bounds.R`	seconds	feasibility unchanged; SD-dependent columns change
08	p08	`01-study1-test-procedure.R`	about 0.2 local h	nearly every result table is typed: a rerun means re-narration; `02-mancl-t-reference.R` must follow the new prefix

Paper	Job	Driver	Stored cost	Notes
09	p09	`20-paired-dropout.R`	about 1.6 local h (500 null replicates; 2.2 at the current default of 1,000)	full published set, not only BR SD 8 (D3); -2.25 typed in 15 places; the informative-dropout mechanism may weaken (inferred)
02, 09	cr2-p09	`cr2-olbdc-check`, p09 setting	about 0.4 local h	follows `20-paired-dropout.R`; its `iso` setting edits `rp` with data.table syntax, which breaks on a tibble; same host as p09's iso run

Not rerun (recommendation; Section 5, D2)

Runs the manuscripts already describe as superseded, withdrawn or for the record: paper 03's pilot (`quick-sim-driver-parallel.R`) and old production (`10-study5-production.R`); paper 06's June Study 2 (contamination) and Study 3 (recovery) and its prototype; paper 09's prototype (`09-dropout-prototype-10min.R`, Section `sec-old`). They stay as stored and are labeled in the text as raw-fixture runs. Paper 10's pre-correction (2026-05-20) outputs and paper 15's repair-artifact cells (prognostic strength 0.6) stay as labeled historical simcore2026 exhibits (D4, D5).

4. Findings outside the parameter error

- **Paper 12 (verified).** It reads a July 2026 run of the paper 11 factorial; the current stored summary no longer has the N = 35 rows, the E1 specification or `sd_beta`, so `report.Rmd` lines 322-327 describe a file that no longer exists in that form, and the July run predates the October DGP changes (inferred). A rerun of the paper 11 factorial cells paper 12 needs is a provenance repair, not a parameter repair; it is listed as optional.

- **Paper 13 (verified).** `report.Rmd` line 70 reads `analysis/data/13-two-component-grid.rds`, which does not exist; the driver writes `13-drop-pb-grid.rds`. Paper 13 renders with every simulated cell missing. A one-line path fix; no simulation.
- **Paper 06 (verified).** `run-study-2-feasible.R` ran 11 workers in an 8-core job, and the report's core-hours use 11.
- **`docs/70` (verified).** It names `nof1power` as paper 04's engine (it is paper 05's), lists `00-common.R` as behind paper 09's results (it is an unrun scaffold), describes a `SIGMA_BR` axis for paper 09 (it is a scalar), and lists `09-calibration-substudy.R` as feeding paper 03 (its output is not read).

5. Decisions (recommended defaults in force unless changed)

- **D1. Biomarker scale.** Standardize the biomarker by the active set's mean and SD before every analysis, as `prepare_long_data()` now does. Tests are unchanged (the map is affine), the true slope stays $-c_{bm}\sigma_{BR}$ per SD (-3.6 at $c_{bm} = 0.45$), and the raw biomarker (mean 124) cannot make the interaction nearly collinear with the drug term. The alternative is to keep the raw scale and report -0.234 per mm Hg.
- **D2. Superseded and withdrawn runs.** Do not rerun; label them as raw-fixture runs (Section 3).
- **D3. Paper 09.** Move to the full published set, not only BR SD 8, so all papers share one parameter table.
- **D4. Paper 10's pre-correction comparison.** Keep the 2026-05-20 runs as the historical arm, labeled simcore2026, and rerun only the refined arm; say that the comparison mixes the DGP correction with the parameter change, or rerun the historical arm under published as well (about 1.8 more zeus-h).
- **D5. Paper 15's repair-artifact cells.** Keep as a labeled historical exhibit; under today's stop-on-non-PD default they would only fail. Rerun the feasible prognostic cells under published.
- **D6. Paper 06's design axes.** Keep the m_PB and m_TV grids and the drivers' structure, adopt the full published table (TV rate 0.35, PB max 6.51) as the base, and report that the axes are now weaker relative to the noise.
- **D7. Paper 03's alternative block.** Time it with `--time` under published before submitting; if it projects above about 24 hours on any host, split it by cell.

6. Host assignment

Pairing rules: job 27 with the E0 isolation, and gap-fill blocks A and B, each on one host (paper 02); drivers 28 and 29 on one host (shared seeds); the nine

paper 10 drivers in one job (the wrapper); the cr2-olbdc `iso` run with paper 09's p09 setting on one host. Everything else is unpaired.

Wave 1 (ready now or after Tier B). Expected hours on each host (zeus-hours divided by the host's relative throughput):

Host	Queue, in order	Hours
zeus (8 cores, 24 GB)	p01-pf, p02-tier1, p02-e0iso, p02-gf-dropout, p02-gf-size, p02-gf-B	about 36
athena (8 cores, 22 GB)	p02-ref-A, p02-factorial, p02-scen, p02-ref-S, p02-ref-N, p15-refT, p10, p15-stress, p15-re	about 25
mercury (6 cores, 12 GB)	check job, p01-n80, p15-gapC	about 28
joe (8 cores, 18 GB)	p03-null, then p03-alt after its timing run	about 4 plus p03-alt

Wave 2 (after Tier C code and its smoke tests). joe takes paper 09 with both cr2-olbdc settings (one host) and paper 06 Study 4 and the bounds; julia takes paper 08 overnight (6 cores, 10 GB; julia is the working machine, so it runs only small, unpaired jobs submitted in the evening); Study 1 and corrected Study 2 go to whichever of zeus and athena frees first. Each Tier C driver is timed on its host before submission, because only local timings exist.

With every queue full, wave 1 ends in about a day and a half; wave 2 adds about a day once its code is ready.

7. Before any submission

1. **Code.** Tier B changes (Section 3), then the commit: the remediation files of `docs/73` Section 5 plus the Tier B drivers. Other uncommitted work stays out; every command uses `--commit HEAD`. Tier C changes are a second commit before wave 2; wave 1 jobs keep their commit.
2. **Checks at the commit.** `42-check-param-sets.R`; a smoke run of every wave 1 driver under the published set; for Tier B drivers, a check that a rerun into the stored folder is refused.
3. **Image.** Unchanged on all four hosts (`...:10504a44fb8d`); athena and joe hold the registry copy built from the same `Dockerfile` and `renv.lock`.
4. **Hosts.** athena and joe plugged in, logged in, lid open; Docker Desktop's Resource Saver off. Mercury's one-minute check job first.

8. Commands, wave 1

Paths are set once; every simrun option is written out (zsh does not split an unquoted variable into words).

```
D=analysis/scripts/carryover-sensitivity
D19=analysis/scripts/quick-sim/carryover-closed-form
D19=$D19/19-g9-planning-and-fig4b.R
P=analysis/data/carryover-sensitivity/published
```

zeus (`docs/73` Section 6 gives the same commands):

```
simrun submit --commit HEAD --cores 8 --mem 24 --label p01-pf -- \
  Rscript $D19 --params published --specs E7null_cr2,E7cr2 --block all
simrun submit --commit HEAD --cores 8 --mem 24 --label p02-tier1 -- \
  Rscript $D/27-run-tier1-g-vs-f-e0.R --params published \
  --reps 500 --n-boot 1 --out-dir $P/tier1-g-vs-f-e0
simrun submit --commit HEAD --cores 8 --mem 24 --label p02-e0iso -- \
  Rscript $D/38-run-e0-isolation.R --params published --reps 1000 \
  --out-dir $P/e0-isolation
simrun submit --commit HEAD --cores 8 --mem 24 --label p02-gf-dropout -- \
  Rscript $D/32-run-g9-gap-fill.R --params published --block A \
  --sub A1-dropout --dropout-per both --out-dir $P/g9-gap-fill/dropout
simrun submit --commit HEAD --cores 8 --mem 24 --label p02-gf-size -- \
  Rscript $D/32-run-g9-gap-fill.R --params published --block A \
  --sub A2-size --out-dir $P/g9-gap-fill/size
simrun submit --commit HEAD --cores 8 --mem 24 --label p02-gf-B -- \
  Rscript $D/32-run-g9-gap-fill.R --params published --block B \
  --out-dir $P/g9-gap-fill/B
```

athena:

```
simrun --host athena submit --commit HEAD --cores 8 --mem 22 \
  --label p02-ref-A -- \
  Rscript $D/36-run-referee-variants.R --params published --block A \
  --out-dir $P/referee-variants/block-A
simrun --host athena submit --commit HEAD --cores 8 --mem 22 \
  --label p02-factorial -- \
  Rscript $D/01-run-factorial.R --params published --arch mvn \
  --out-dir $P/factorial
simrun --host athena submit --commit HEAD --cores 8 --mem 22 \
  --label p02-scen -- \
  Rscript $D/06-g-vs-f-comparison.R --params published \
  --reps 500 --n-boot 1 --cells paper02 --out-dir $P/g-vs-f-comparison
```

```
simrun --host athena submit --commit HEAD --cores 8 --mem 22 \
  --label p02-ref-S -- \
  Rscript $D/36-run-referee-variants.R --params published --block S \
  --out-dir $P/referee-variants/block-S
simrun --host athena submit --commit HEAD --cores 8 --mem 22 \
  --label p02-ref-N -- \
  Rscript $D/36-run-referee-variants.R --params published --block N \
  --out-dir $P/referee-variants/block-N
simrun --host athena submit --commit HEAD --cores 8 --mem 22 \
  --label p15-refT -- \
  Rscript $D/36-run-referee-variants.R --params published --block T \
  --out-dir $P/referee-variants/block-T
simrun --host athena submit --commit HEAD --cores 8 --mem 22 \
  --label p10 -- \
  env PMSIM_PARAMS=published bash $D/run-paper10-refined.sh
simrun --host athena submit --commit HEAD --cores 8 --mem 22 \
  --label p15-stress -- \
  Rscript $D/28-run-g9-prognostic-stress.R --params published \
  --out-dir $P/g9-prognostic-stress
simrun --host athena submit --commit HEAD --cores 8 --mem 22 \
  --label p15-re -- \
  Rscript $D/29-run-g9-random-effects.R --params published \
  --out-dir $P/g9-random-effects
```

mercury (after the check job of `docs/73` Section 5):

```
simrun --host sim-mercury-ts submit --commit HEAD --cores 6 --mem 12 \
  --label p01-n80 -- \
  Rscript $D19 --params published --specs E7null_cr2,E7cr2 --block N
simrun --host sim-mercury-ts submit --commit HEAD --cores 6 --mem 12 \
  --label p15-gapC -- \
  Rscript $D/32-run-g9-gap-fill.R --params published --block C \
  --out-dir $P/g9-gap-fill/C
```

joe:

```
D3=analysis/scripts/latent-class-mixture-application
P3=analysis/data/latent-class-mixture-application/published
simrun --host joe submit --commit HEAD --cores 8 --mem 18 \
  --label p03-null -- \
  Rscript $D3/12-referee-rerun.R --params published --block null \
  --out-dir $P3/referee-rerun-null
```

The paper 03 command's block argument and the paper 10 and 15 commands take their final form with the Tier B changes; check each driver's usage line before submitting.

9. Monitoring and failure handling

Zeus and mercury ignore `--time`, and zeus has stalled before (jobs 20, 39, 47 and 52 ran 125 to 263 hours on work that takes hours). Each job is checked against its expected hours (Section 6, or the timing run for Tier C) and cancelled at twice that. A Mac job stops if the Mac sleeps or Docker Desktop quits; it then shows as failed, and is resubmitted.

10. After the runs, per paper

- **01:** render; every [pending] marker disappears.
- **02:** `docs/71` Section 8 (repoint paths, assemble the gap-fill jobs, ladder, referee reanalysis, figures, Methods parameter table, new Results).
- **03:** copy the referee rerun into the folder the report reads (the null block from job 38 was never copied, verified), relabel the superseded runs, re-narrate.
- **06, 08, 09:** replace typed numbers with values read from the new outputs, then re-narrate; the -2.25 true value and the fixture SDs disappear from the text.
- **10:** point the Rmd at `output/paper10-published/`; label the historical arm.
- **15:** create the paper folder (split plan step 5) and read the published-set outputs.

11. Evidence status

- **Measured:** the host benchmarks of Section 2; the stored jobs' durations; the image tag on every host.
- **Verified (read in code, provenance or stored files):** each driver's parameter source, output path and the files each manuscript reads; the findings of Section 4.
- **Inferred:** joe's and mercury's 8-worker throughput; the G10 overhead; costs given as local hours; the scientific consequences flagged in Section 3 (weaker contamination signal relative to noise, weaker informative dropout).
- **Not done:** the Tier B and Tier C code changes; the paper 03 alternative block's timing.